

Yifan Yang

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RESEARCH INTERESTS

I am interested in understanding and designing practical machine learning methods from a theoretical perspective. Currently, I focus on adaptive and parameter-free *non-convex optimization* methods and *efficient attention mechanisms*, particularly linear attention based on test-time training (TTT).

I aim to develop a framework with strong theoretical foundations that is efficient and easy to apply. I believe that general principles underlie the behavior of machine learning methods. Uncovering these principles may require an approach closer to physics, which seeks patterns amid noise and complexity, than to purely mathematical analysis. Influenced by "The Bitter Lesson", I view efficiency as a key factor in machine learning. These perspectives guide my efforts to develop methods that are *theoretically sound*, *computationally efficient*, and *useful in practice*.

EDUCATION

Nanjing University September 2026 - present
M.Sc. student, LAMDA Group, School of Artificial Intelligence

- Major: Computer Science and Technology
- Advisor: Asst. Prof. Peng Zhao

Harbin Institute of Technology August 2022 - June 2026
B.Eng., School of Astronautics

- Major: Automation
- GPA: 94.536/100; rank: 4/108

HONORS / AWARDS

The 48th Annual World Finals of the ICPC - Rank 24 / 141	September 2024
The 2023 ICPC Asia-East Continent Final Contest - Gold Medal, Rank 6 / 280	January 2024
The 2023 ICPC Asia Nanjing Regional Contest - Gold Medal	November 2023
The 2023 ICPC Asia Shenyang Regional Contest - Gold Medal	November 2023
National Scholarship	December 2023
The 2022-2023 ICPC Asia Hong Kong Regional Contest - Silver Medal	January 2023
CCF Certified Software Professional, Rank 2 / 6616	May 2025

PROJECTS / RESEARCH EXPERIENCE

Linear Attention from the Online Learning Perspective October 2025 - present
LAMDA Group, School of Artificial Intelligence, Nanjing University

- On memory-intensive tasks such as associative recall, TTT-based linear attention methods outperform traditional linear attention methods. Each layer performs data-dependent compression of a stream of key-value pairs. This view enables a theoretical analysis of their memory capabilities through the lens of online learning.

- **Method design.** Explore connections between TTT-based linear attention and online learning theory using the mean squared error (MSE) loss and online gradient descent in the standard DeltaNet formulation. From this perspective, DeltaNet can be extended to Conceptual DeltaNet using conceptual online gradient descent.
- **Experiments.** Compare models with softmax attention, Gated Linear Attention, DeltaNet, Conceptual DeltaNet, and Gated DeltaNet. Train these models using *Triton kernels* and a specialized pretraining framework for efficiency, and evaluate them on both *synthetic* and *real-world* tasks. **The project is ongoing.**

SKILLS

- Programming languages: C/C++, Python, and shell scripting.
- Algorithms and data structures: expertise in
 - Basic algorithms: binary search, greedy algorithms, dynamic programming, network flow, etc.
 - Advanced algorithms and data structures: chunk algorithm, k-d trees, segment trees, splay trees, etc.
- Machine learning tools: PyTorch, Triton, and TileLang.